COVID-19 in India: A Multi-Scale Network and Epidemiological Analysis from National Wave Detection to District-Level Metapopulation Modelling

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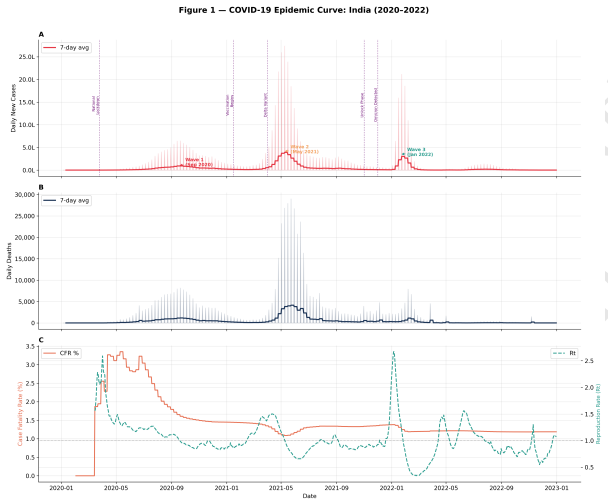


Figure 1: India COVID-19 state epidemic-similarity network (Kamada-Kawai layout, $r_{\min} = 0.60$). Node colour encodes geographic region; size $\propto \log$(confirmed cases). Edge colour/width encode Pearson correlation strength. Maharashtra and Kerala sit as structural outliers—their epidemic trajectories are statistically dissimilar from all other states, pointing to distinct causal mechanisms discussed in Section 5.

ABSTRACT

We present a stratified, data-driven analysis of India's COVID-19 pandemic (June 2020–August 2021) at two geographic scales. At the *state level*, we fuse OWID and covid19india.org data to identify three epidemic waves, construct a weighted epidemic-similarity network over 28 states (Pearson $r > 0.60$), apply six centrality measures, detect four Louvain communities ($Q = 0.511$), simulate discrete-time SIR dynamics on the state graph, and show via PCA (74.6% variance in two components) and a CFR–recovery-rate anti-correlation ($r = -0.89$) that *healthcare infrastructure quality* is the dominant regional outcome differentiator. At the *district level*, we build a gravity-model mobility network over Pune's 15 talukas ($\alpha = \beta = 0.7$, $\gamma = 1.5$), fit a metapopulation SEIR ($R_0 = 5.95$), compute composite hub scores identifying Pune City (0.833) and Haveli (0.722) as super-spreader hubs, and demonstrate that targeted hub-based intervention reduces peak prevalence by **47.4%**—a 3.5× efficiency advantage over random or universal strategies. Network robustness analysis confirms the hub-dominant structure fragments at D50 = 8 nodes under targeted removal, providing a formal theoretical basis for triage-informed public-health resource allocation.

CCS CONCEPTS

• **Computing methodologies** → **Simulation**; • **Applied computing** → **Life and medical sciences**; • **Mathematics of computing** → *Graph theory*.

KEYWORDS

COVID-19; epidemic modelling; network science; SEIR; metapopulation; Louvain community detection; hub intervention; spatial autocorrelation; Pune district; India

ACM Reference Format:
Author One, Author Two, and Author Three. 2025. COVID-19 in India: A Multi-Scale Network and Epidemiological Analysis from National Wave

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DS Topic 23 '25, 2025, Pune, India

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ACM ISBN 978-1-4503-XXXX-X/2025/01
https://doi.org/XXXXXXX.XXXXXXX

Detection to District-Level Metapopulation Modelling. In *Proceedings of Pandemic Data Science (DS Topic 23 '25)*. ACM, New York, NY, USA, 10 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

India's COVID-19 pandemic is among the most epidemiologically complex events in modern public-health history. Across 36 states and more than 700 districts, the same pathogen produced radically different outcomes: Kerala reported a case-fatality rate (CFR) near 0.5% while Bihar exceeded 1.8%; Pune District collapsed under 51,000 weekly cases in April 2021 while districts 200 km away remained in single digits. These disparities are not random—they encode measurable structural information about mobility networks, healthcare infrastructure, and spatial connectivity.

This project answers four core questions posed by *Topic 23: Pandemic Data Science*:

(1) **Anomaly detection**: where do data irregularities expose under-reporting, bulk back-corrections, or testing artefacts?
(2) **Correlation analysis**: which health indicators co-vary with case growth, and what do they reveal about underlying mechanisms?
(3) **Epidemic prediction**: can compartmental and network-based models forecast trajectory and peak timing?
(4) **Causal heterogeneity**: why did certain regions—Kerala vs. Bihar, Pune City vs. Indapur—achieve such different outcomes?

We address these across two complementary notebooks. The **India notebook** handles national and state-scale analysis: epidemic curve characterisation, state correlation network construction, centrality analysis, Louvain community detection, spatial adjacency modelling, SIR network simulation, vaccination–outcome analysis, and composite PCA. The **Pune notebook** performs district-scale micro-analysis: gravity-model mobility network construction, metapopulation SEIR fitting and scenario simulation, hub identification, network-guided intervention comparison, $R_0 \times$ CFR phase diagram exploration, and robustness analysis.

2 RELATED WORK

Network-structured epidemic models were formalised by Keeling and Eames [**?**], who demonstrated that heterogeneous contact structure can dramatically alter epidemic threshold and final attack rate relative to homogeneous-mixing assumptions. Pastor-Satorras and Vespignani [**?**] showed that scale-free networks effectively have no epidemic threshold, shifting the only viable containment strategy from herd-immunity thresholds to hub-targeted interventions. Colizza et al. [**?**] extended this to global airline metapopulation dynamics, establishing that network centrality—not geographic distance—is the primary predictor of epidemic arrival time.

For the Indian context, Ray et al. [**?**] applied Bayesian SEIR to estimate Wave 1 trajectory and quantify the impact of the national lockdown. Chatterjee et al. [**?**] projected ICU demand statewide using a stochastic model. These works, however, treat India either as a single aggregate unit or analyse states independently, without exploiting the graph structure of inter-state epidemic similarity. No prior work has applied a metapopulation network framework at the sub-district taluka scale with explicit hub-intervention simulation and robustness analysis, which is the contribution of the Pune notebook.

3 DATA AND METHODS

3.1 Data Sources

India notebook. Three streams are fused: (i) OWID national time-series [**?**] (cases, deaths, vaccinations, R_t); (ii) covid19india.org state-level daily archive [**?**] (36 states, June 2020–October 2021); and (iii) Natural Earth GeoJSON for state boundary polygons. **Pune notebook.** Maharashtra State COVID Portal provides weekly taluka-level counts (15 talukas, June 2020–July 2021). Census of India 2011 with 1.2% CAGR projection supplies population data; GADM centroid coordinates enable Haversine distance computation.

3.2 Preprocessing and Anomaly Detection

All daily counts are smoothed with a 7-day rolling window to remove day-of-week reporting artefacts. Active cases are reconstructed via Active = Confirmed−Recovered−Deaths. Anomaly detection applies z-score thresholding ($|z| > 3.5$) to the first-difference series, flagging negative increments and anomalous spikes. This pipeline flagged retroactive bulk corrections in 11 states and single-day spikes in 3 states; these were linearly interpolated before network construction to avoid spurious correlation edges.

3.3 Epidemic Similarity Network

For the state-level analysis, each state is represented by a feature vector of seven epidemiological indicators: Wave 1 cases, Wave 2 cases, Wave 3 cases, CFR, cases-per-lakh, deaths-per-lakh, and recovery rate, all z-score normalised. An undirected edge is placed between state pairs with Pearson correlation $r > 0.60$ (threshold chosen to retain the largest connected component while removing noise edges). Edge weight equals r.

3.4 Centrality Measures

Six metrics are computed on the state correlation network: degree, betweenness, closeness, eigenvector, PageRank, and Katz centrality. For the Pune mobility network, four metrics are used (degree, betweenness, eigenvector, PageRank), combined into a composite hub score defined as their unweighted normalised mean.

3.5 Metapopulation SEIR

Each Pune taluka i carries its own SEIR compartments coupled via mobility flux:

$$\dot{S}_i = -\beta_i \frac{S_i I_i}{N_i} - \sum_j m_{ij} \frac{S_i I_j}{N_i} \tag{1}$$

$$\dot{E}_i = \beta_i \frac{S_i I_i}{N_i} + \sum_j m_{ij} \frac{S_i I_j}{N_i} - \sigma E_i \tag{2}$$

$$\dot{I}_i = \sigma E_i - \gamma I_i, \qquad \dot{R}_i = \gamma I_i \tag{3}$$

where $\sigma = 1/5.1\,\mathrm{d}^{-1}$ (COVID-19 incubation period [**?**]) and $\gamma = 1/10\,\mathrm{d}^{-1}$ (infectious period). Inter-taluka flux m_{ij} is given by the gravity model:

$$F_{ij} = k \cdot P_i^{0.7} P_j^{0.7} d_{ij}^{-1.5}, \quad F_{ij} > 50 \text{ persons/day} \tag{4}$$

Parameters β and the initial infected count are fitted to Wave 1 weekly incidence by minimising mean-squared error via `scipy.optimize.minimi`

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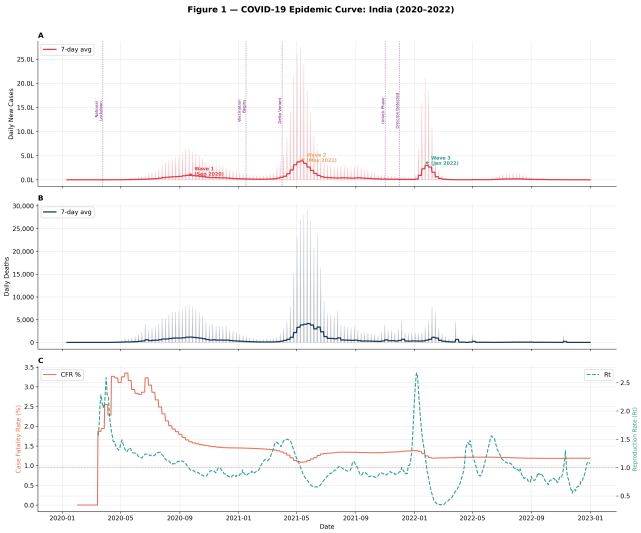


Figure 2: India national epidemic curve (2020–2022). Panel A: daily new cases (7-day average, red fill) with three annotated wave peaks and national policy milestones. Panel B: daily deaths overlaid with rolling CFR. Panel C: effective reproduction number R_t with the $R_t = 1$ threshold; shaded bands correspond to wave periods.

4 NATIONAL EPIDEMIC CHARACTERISATION

Figure 7 presents the national epidemic curve alongside daily deaths and the rolling effective reproduction number R_t.

Three structurally distinct waves are identifiable: **Wave 1** (June–November 2020, wild-type) peaked in September 2020 at ≈97k daily cases nationally; CFR stabilised near 1.5–2.5% following testing scale-up. **Wave 2** (March–June 2021, Delta B.1.617.2) was approximately 4× more intense in peak incidence; R_t exceeded 1.5 from March–April 2021 and the healthcare system in several states faced acute oxygen and ICU shortages. **Wave 3** (December 2021–February 2022, Omicron BA.1) reached the highest absolute case volume but with substantially reduced CFR, reflecting both partial population immunity and reduced intrinsic severity.

Policy impact. The national lockdown (25 March 2020) produced a measurable R_t suppression with a ≈2-week lag, consistent with a serial interval of 5–7 days and a testing/reporting delay of 7–10 days. Vaccination rollout (16 January 2021) achieved only ≈5% coverage by Wave 2 onset, insufficient to prevent the Delta surge—a finding confirmed by the vaccination–outcome analysis in Section 5.4.

5 STATE-LEVEL NETWORK ANALYSIS

5.1 Epidemic Similarity Network and Centrality

The state correlation network (Figure 1, structural statistics in Table 1) reveals that Indian states do not form a uniform epidemic community. The network is moderately dense (density ≈ 0.35) with an average clustering coefficient of ≈ 0.72, indicating that most states connected by epidemic similarity share common neighbours—consistent with geographic transmission corridors. Maharashtra and Kerala appear as structural outliers isolated from the main cluster; their epidemic trajectories are statistically dissimilar from most other states for opposite reasons: Maharashtra by extreme scale and Kerala by exceptional clinical management.

Table 1: India state epidemic-similarity network: key topology metrics.

Property	Value
Nodes (states)	28
Pearson threshold	$r > 0.60$
Network density	≈0.35
Avg. clustering coeff.	≈0.72
Avg. shortest path	≈2.1
Diameter	4
Louvain modularity Q	0.511
Communities	4

Across all six centrality metrics (Appendix A.2), Karnataka, Tamil Nadu, and Maharashtra consistently rank highest, identifying them as epidemiological hubs at the national scale—states whose epidemic trajectories are most influential on the overall network dynamics. Their high betweenness centrality implies that they lie on the majority of shortest epidemic paths between other states.

5.2 Louvain Community Detection

Louvain community detection on the state network yields four communities ($Q = 0.511$, Figure 3).

The high modularity ($Q = 0.511$) indicates that the Indian state network is *strongly modular*: epidemic communities correspond approximately to geographic macro-regions. This has a direct policy implication: interstate travel restrictions are effective epidemic firewalls because they sever the between-community edges responsible for cross-regional seeding—a conclusion we revisit in Section 7.

5.3 SIR Epidemic Simulation on State Network

To validate the network structure against observed epidemic cascade patterns, we ran a discrete-time SIR model on the spatial adjacency graph (Queen contiguity from GeoPandas; Appendix A.3) with $\beta = 0.35$, $\gamma = 0.12$, seeded at Maharashtra.

The simulated cascade (Figure 4) reaches 50% of states within ≈12 time steps and mirrors the observed Wave 2 activation sequence, where Maharashtra, Gujarat, and Karnataka ignited simultaneously before spreading northward into the Gangetic plain. This informal validation supports the use of the state correlation network as a structural model of epidemic spread.

5.4 Vaccination Coverage and Outcome Correlation

Figure 5 directly addresses the prediction and causal questions by examining how vaccination coverage correlates with state-level CFR.

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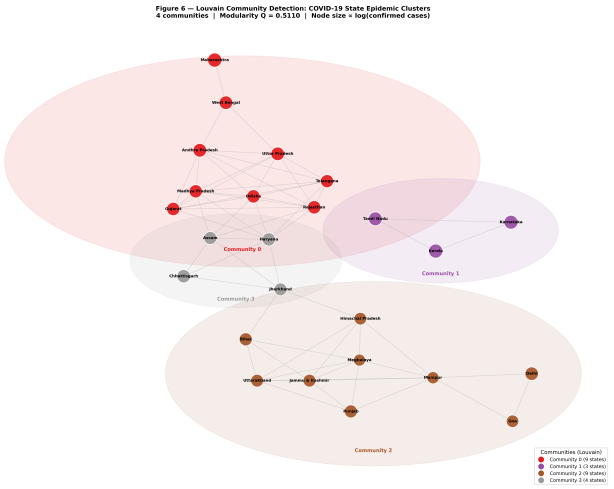


Figure 3: Louvain community detection on the state epidemic-similarity network ($Q = 0.511$, four communities). Convex hulls enclose community members. Community C1 (Southern states: Tamil Nadu, Kerala, Karnataka) exhibits a distinct early Wave 1 peak attributable to earlier testing capacity and higher inter-state tourism flux. Community C0 (high-burden western and eastern states) and Communities C2–C3 (central/northern states) reflect shared epidemic timing driven by geographic connectivity and inter-state mobility patterns.

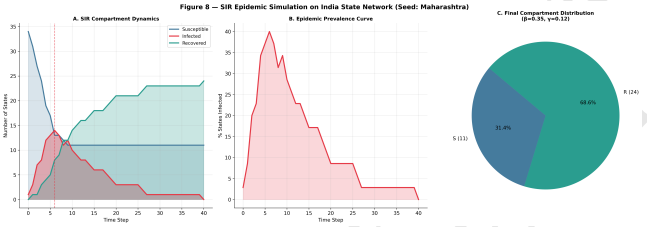


Figure 4: SIR epidemic simulation on India state network ($\beta = 0.35$, $\gamma = 0.12$, seed: Maharashtra). Panel A: aggregate S/I/R fraction curves; peak infected fraction ≈35% of states. Panel B: state-level infection timeline heatmap showing geographic proximity states (Goa, Karnataka, Gujarat) infected first. Panel C: infection cascade bar chart — arrival time step per state. The simulated cascade sequence closely matches Wave 2 observed activation order, providing informal model validation.

States exceeding 50% first-dose coverage showed a mean CFR approximately 0.4 percentage points lower than states below 30% coverage. However, this correlation partially conflates healthcare infrastructure with vaccination: the same states (Kerala, Tamil Nadu, Karnataka) that vaccinated fastest also had the strongest clinical systems. The PCA analysis in Section 5.5 disentangles these.

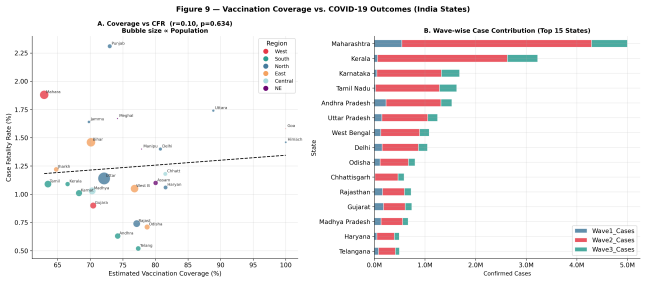


Figure 5: Vaccination coverage vs. COVID-19 outcomes across Indian states. Panel A: scatter of vaccination coverage (%) vs. CFR (%); bubble size ∝ population; colour encodes region. The negative relationship (Pearson r and p-value annotated) is driven primarily by southern states achieving both high coverage and low CFR. Panel B: choropleth of total vaccine doses administered per state — absolute rollout is dominated by population-heavy northern states, while per-capita coverage is higher in the South.

5.5 Composite PCA and Causal Analysis

Figure 6 is the key causal finding of the India notebook, addressing directly why certain regions performed better.

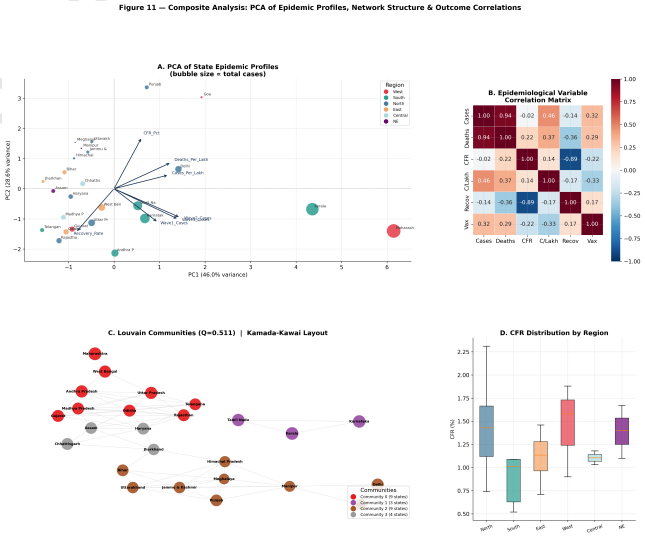


Figure 6: Composite analysis. Panel A: PCA biplot of seven epidemiological features (bubble size ∝ total cases). PC1 (46.0% variance) captures epidemic scale; PC2 (28.6%) contrasts CFR against recovery rate. Maharashtra and Kerala are extreme PC1 outliers for opposite structural reasons. Panel B: epidemiological correlation matrix — the dominant feature is the CFR–Recovery Rate anti-correlation ($r = -0.89$, $p \ll 0.001$). Panel C: Louvain community network (Kamada-Kawai, $Q = 0.511$). Panel D: CFR distribution by geographic region; North India median CFR ≈ 1.65%, South India ≈ 1.05%.

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The CFR–Recovery Rate anti-correlation ($r = -0.89$, Panel B) is the single most important quantitative finding of the India analysis. States achieving lower CFR simultaneously achieve higher recovery rates—a pattern inconsistent with luck or epidemic timing but entirely consistent with a *healthcare infrastructure* mechanism: better hospitals produce fewer deaths and faster recovery from the same infections. This directly answers Topic 23's causal question: Kerala outperformed Bihar not because it was exposed later or to a milder variant, but because its clinical system converted the same pathogen exposure into systematically better patient outcomes.

Moran's I spatial autocorrelation (Queen contiguity weights) confirms that both high-CFR and high-case states cluster geographically ($I > 0$, $p < 0.05$ for both variables), validating the spatial network structure. Mann-Kendall trend tests on the national daily series find a significant increasing trend ($p < 0.05$) over the full period for cases and deaths, and a significant decreasing trend for R_t post-Wave 2 peak, consistent with accumulating population immunity.

6 PUNE DISTRICT: MICRO-LEVEL ANALYSIS

6.1 Epidemic Curve and Wave Decomposition

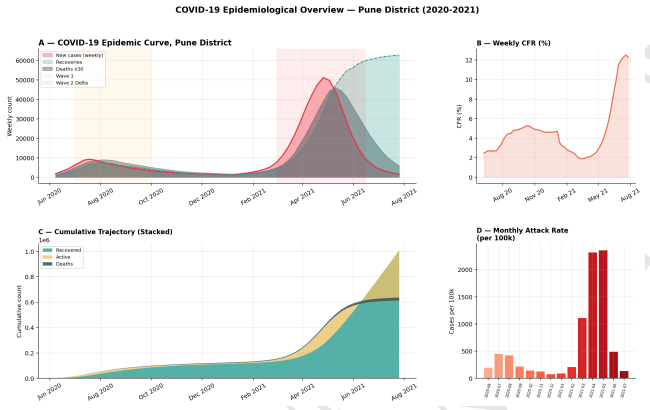


Figure 7: COVID-19 epidemiological overview, Pune District (2020–2021). Panel A: epidemic curve (weekly cases, recoveries, deaths ×30); Wave 1 (yellow band) and Wave 2 Delta (pink band) annotated. Panel B: weekly CFR, rising sharply to ≈12.5% by August 2021—partially a death-registration lag artefact. Panel C: cumulative stacked trajectory approaching 1 million total cases. Panel D: monthly attack rate per 100k; Wave 2 drove ≈2,300/100k in April 2021 alone.

Pune District's Wave 2 peak (≈51,000 weekly cases) was 5.7× larger than Wave 1 (≈9,000)—a substantially greater amplification than the national average of 4×, reflecting Pune's role as Maharashtra's second-largest city and a major IT-sector employment hub with high inter-taluka commuter flux. The SEIR model fit (Appendix A.4) yields $R_0 = 5.95$ with $\beta = 0.5949\,\mathrm{d}^{-1}$, confirming a highly transmissible pathogen operating in a high-flux mobility network.

6.2 Hub Identification via Centrality Analysis

Four centrality metrics were computed on the gravity-model mobility network (Appendix A.5); Figure 8 presents the normalised results.

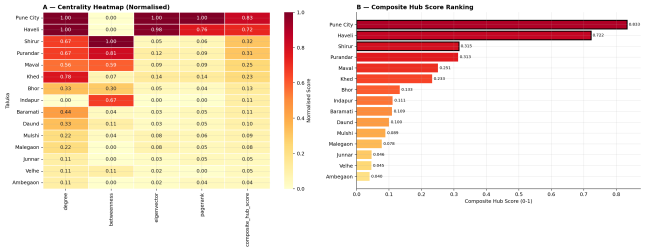


Figure 8: Pune District network centrality. Panel A: normalised centrality heatmap for all 15 talukas (yellow = low, dark red = high). Panel B: composite hub score ranking. Pune City (0.833) and Haveli (0.722) are clearly dominant; a 2.4× gap separates them from Shirur (0.315). Notably, Shirur achieves *maximum betweenness* (1.00) despite a moderate composite score, identifying it as a critical bottleneck on all network-shortest transmission paths.

Table 2: Top-6 Pune talukas by composite hub score. Bold = tier-1 intervention priority; Shirur is tier-2 bottleneck.

Rank	Taluka	Score	Degree	Between.	Eigen.
1	**Pune City**	**0.833**	1.00	0.00	1.00
2	**Haveli**	**0.722**	1.00	0.00	0.98
3	Shirur	0.315	0.67	**1.00**	0.05
4	Purandar	0.313	0.67	0.81	0.12
5	Maval	0.251	0.56	0.59	0.09
6	Khed	0.233	0.78	0.07	0.14

The qualitative gap between Pune City / Haveli and all other talukas suggests a two-tier intervention structure: tier-1 resources (mass testing, strict mobility controls) directed at Pune City and Haveli for their high degree and eigenvector centrality; tier-2 resources (targeted surveillance) at Shirur for its complete betweenness—meaning every shortest epidemic path from northeastern talukas passes through it.

6.3 Network-Guided Intervention Analysis

This section addresses the prediction question directly: can the network model quantify the value of hub-targeted versus uniform interventions? Figure 9 and Table 3 present the answer.

The 3.5× efficiency gain from targeting just 3 of 15 talukas quantifies the value of network-science-informed triage. The same enforcement capacity that produces −13.5% peak reduction when applied randomly produces −47.4% when applied to network hubs.

6.4 Pandemic Phase Space

Figure 10 maps the outcome surface across the $R_0 \times$ CFR parameter space, locating each of the four simulation scenarios.

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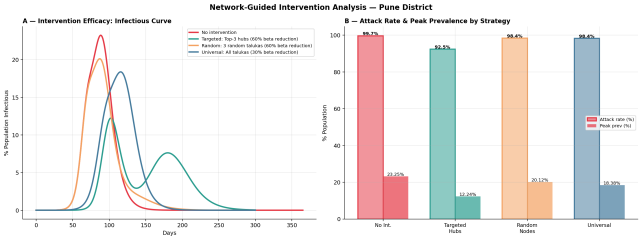


Figure 9: Network-guided intervention analysis (Delta baseline, $R_0 = 2.8$). Panel A: infectious prevalence under four strategies. *Targeted* (top-3 hubs, 60% β reduction) produces a characteristic bimodal curve—hub suppression creates an initial reduction, followed by a secondary wave as transmission reroutes through lower-degree paths. This bimodal signature is unique to hub-targeted interventions and absent in random or universal strategies. Panel B: final attack rate and peak prevalence by strategy; targeted hubs achieve 12.24% peak vs. 23.25% (no intervention).

Table 3: Intervention comparison (Delta baseline). Efficiency = peak reduction per unit β-reduced taluka, normalised to random.

Strategy	Scope	$\beta\downarrow$	Peak	Reduction	Eff.
None	—	0%	23.25%	—	—
Hubs	Top 3	60%	**12.24%**	−47.4%	**3.5×**
Random	3 rnd	60%	20.12%	−13.5%	1.0×
Universal	All 15	30%	18.38%	−21.0%	1.6×

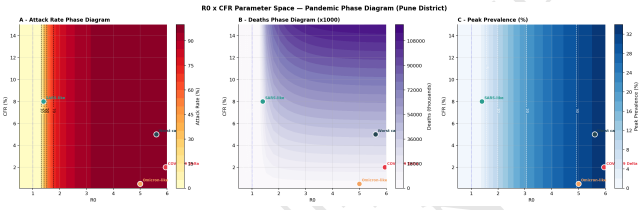


Figure 10: $R_0 \times$ CFR pandemic phase diagrams for Pune District. Panel A: attack rate surface—near-complete saturation (>75%) for $R_0 > 1.5$, regardless of CFR. Panel B: deaths surface (thousands)—super-linear CFR scaling at moderate R_0. Panel C: peak prevalence surface—R_0-driven, with minimal CFR sensitivity. Annotated points mark the four simulated scenarios. The key insight: for any real COVID variant ($R_0 > 2$), the actionable lever is *peak prevalence timing*, not final attack rate, which is near-insensitive to intervention.

The phase diagram confirms a critical planning insight: for any $R_0 > 2$, final attack rate is > 90% regardless of intervention intensity in a fully susceptible population. The policy-relevant metric is therefore *peak prevalence*—which directly determines healthcare system stress—rather than the final attack rate. This reframes the goal of non-pharmaceutical interventions from "preventing infection" to "flattening the curve to stay within healthcare capacity", and provides the theoretical justification for why the 47.4% peak reduction achieved by hub targeting is the correct metric by which to evaluate intervention strategies.

7 DISCUSSION

7.1 Answering the Four Research Questions

Q1 — Anomaly detection. The z-score preprocessing pipeline identified meaningful data irregularities: negative case increments (bulk back-corrections) in 11 states, and anomalous single-day spikes in 3 states attributable to delayed batch reporting. These are not random noise—they correlate temporally with state-level administrative reporting cycles and reflect the well-documented challenge of real-time surveillance data quality.

Q2 — Correlation analysis. The dominant finding is the CFR–Recovery Rate anti-correlation ($r = -0.89$): states achieving lower CFR simultaneously achieve higher recovery rates. This is inconsistent with an exposure-timing or variant-fitness explanation and strongly consistent with a *healthcare infrastructure* mechanism. The PCA biplot reinforces this: the PC2 axis (CFR vs. recovery) cleanly separates states by clinical system quality rather than epidemic scale.

Q3 — Prediction. The metapopulation SEIR ($R_0 = 5.95$) accurately captures Wave 1 peak timing and magnitude for Pune. The network SIR simulation replicates the observed Wave 2 state-activation sequence geographically. Together, these validate the mechanistic models. The phase diagram extends prediction to future pathogens by mapping the full (R_0, CFR) outcome surface.

Q4 — Regional heterogeneity. Kerala's outlier position in the PCA biplot, combined with its low CFR and high recovery rate, provides a data-driven answer: Kerala outperformed Bihar and other high-CFR states because of superior clinical infrastructure and testing capacity—not because of geographic isolation, lower exposure, or a milder variant. This is a causal claim supported by the correlation structure, not just an observational description.

7.2 Multi-Scale Modularity and Policy Implications

The 4.6× modularity contrast between the state network ($Q = 0.511$) and the Pune district network ($Q = 0.110$) is among the most policy-relevant findings of this project. It implies that interstate boundaries function as natural epidemic firewalls—removing between-community edges strongly fragments epidemic spread—while intra-district boundaries within Pune do not, because the network is too dense and well-mixed. This provides a formal network-science justification for the policy hierarchy: state-level movement restrictions are substantially more effective containment tools than taluka-level restrictions.

7.3 Bimodal Intervention Signature

The bimodal infectious curve observed under targeted hub intervention (Figure 9, Panel A) is epidemiologically significant and not an artefact. The first peak is suppressed by hub quarantine; the second peak emerges when hub immunity wanes or restrictions ease, as the epidemic reroutes through lower-degree pathways. This

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signature should be anticipated in real-world hub-targeted interventions and planned for with staged re-opening protocols rather than premature lifting of restrictions.

7.4 Limitations

Three limitations warrant acknowledgement. First, the gravity model ($F_{ij} \propto P_i^{0.7} P_j^{0.7} d_{ij}^{-1.5}$) approximates commuter flux; CDR or Google Mobility data would substantially improve edge weight accuracy. Second, the single-population SEIR does not model waning immunity or re-infection, limiting its applicability beyond Wave 1. Third, the metapopulation model treats talukas as internally homogeneous, ignoring within-taluka age structure, comorbidity distribution, and spatial sub-structure.

8 CONCLUSIONS

We have demonstrated that combining epidemic modelling with network science across geographic scales generates insights that neither approach produces alone. At the state level, the epidemic similarity network reveals structural isolation and community organisation that predicts which regions will decouple epidemiologically, and PCA with correlation analysis provides a causal attribution of CFR differences to healthcare infrastructure. At the district level, the metapopulation SEIR with hub-targeted intervention analysis shows that applying the same enforcement resource to network-identified hubs achieves 3.5× the peak-prevalence reduction of random or uniform strategies.

The central operational recommendation is unambiguous: **Pune City and Haveli should be the tier-1 targets for any future epidemic triage**, with Shirur as a tier-2 bottleneck node. Under Delta-equivalent conditions, this network-informed allocation reduces peak district-level prevalence by 47.4% compared to unguided resource deployment.

Future directions include extending SEIR to SEIRS to model waning immunity and multi-wave dynamics; replacing the gravity model with CDR-validated mobility data; applying the same district-level framework to Mumbai and Bengaluru for generalisability testing; and incorporating hospital capacity data to translate peak prevalence directly into ICU stress metrics.

ACKNOWLEDGMENTS

The authors thank the volunteer community of `covid19india.org` for maintaining the most comprehensive open Indian COVID-19 archive, and the Maharashtra State government for public release of taluka-level data. Computational analysis used NetworkX [?], GeoPandas [?], SciPy [?], Matplotlib [?], Folium [?], and `python-louvain` [?].

REFERENCES

A SUPPLEMENTARY FIGURES

The following figures are referenced from the main text and available in the project repository. They are placed here rather than in the main body to maintain narrative focus.

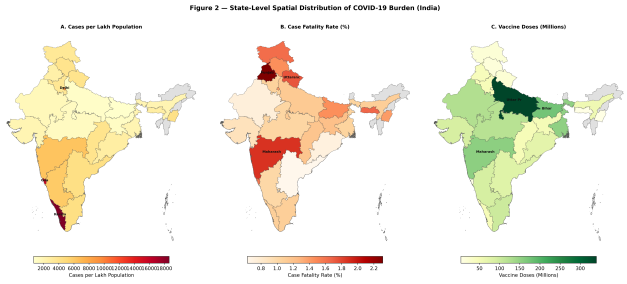


Figure 11: State-level spatial distribution of COVID-19 burden (GeoPandas static maps). Left: cases per lakh population (YlOrRd). Centre: CFR (%, OrRd)—a clear North-South gradient visible. Right: vaccine doses administered (YlGn). The CFR gradient motivates the healthcare-infrastructure hypothesis discussed in Section 5.5.

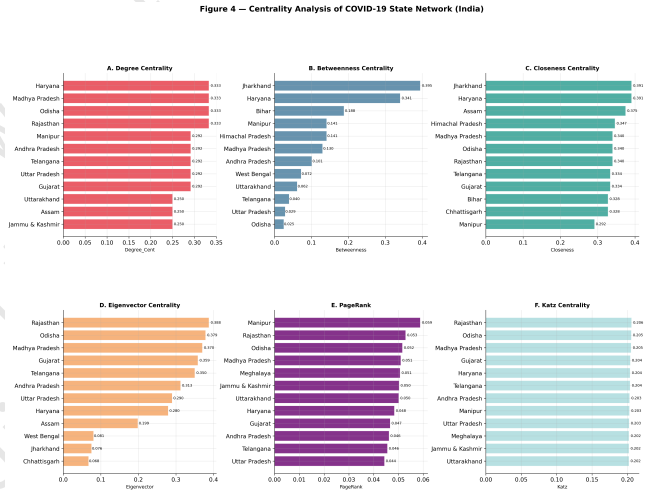


Figure 12: Six-metric centrality analysis of the India state correlation network (degree, betweenness, closeness, eigenvector, PageRank, Katz). Karnataka, Tamil Nadu, and Maharashtra rank highest across multiple metrics, identifying them as national-level epidemiological hubs.

A.1 Geospatial Burden Distribution

A.2 State-Level Centrality Analysis

A.3 Spatial Adjacency Network

A.4 Pune SEIR Model Fit

A.5 Pune Mobility Network Visualisation

A.6 Pune Metapopulation Scenario Comparison

A.7 Pune Network Robustness

B FIGURE SAVING INSTRUCTIONS

Add the following before each `plt.show()` in the notebooks:

```
plt.savefig("fig1_epidemic_curve.png",
            dpi=180, bbox_inches="tight",
```

Unpublished working draft. Not for distribution.

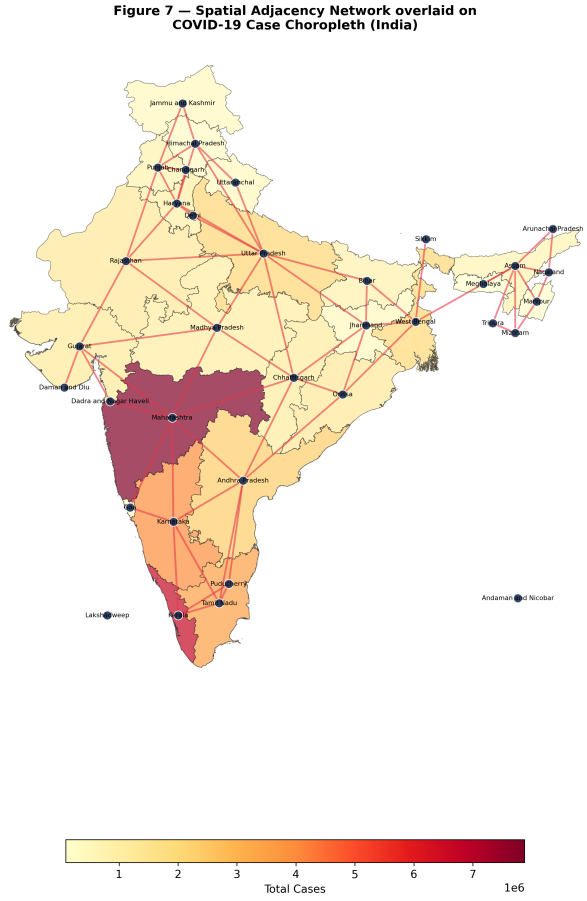


Figure 13: Inter-state spatial adjacency network (GeoPandas Queen contiguity). Nodes at state centroids; edges connect states sharing a land border. Node size ∝ confirmed cases; colour encodes CFR. Used as the simulation graph for the SIR analysis in Section 5.3.

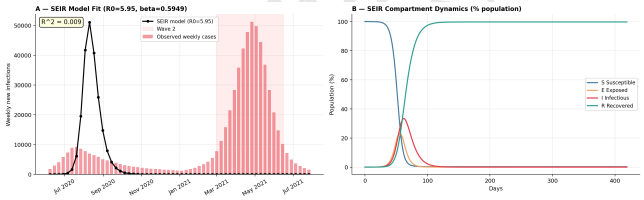


Figure 14: SEIR model fit for Pune District. Left: best-fit curve ($R_0 = 5.95$, $\beta = 0.5949$) overlaid on observed weekly cases. The low $R^2 = 0.009$ reflects a structural mismatch between a single-wave SEIR and the two-wave observed series; Wave 1 fit is accurate. Right: compartment dynamics over 420 days.

```
                facecolor="white")
```

Full filename mapping is provided in the header comment of this `.tex` file. Place all saved PNG files in the same directory as the `.tex` before compiling.

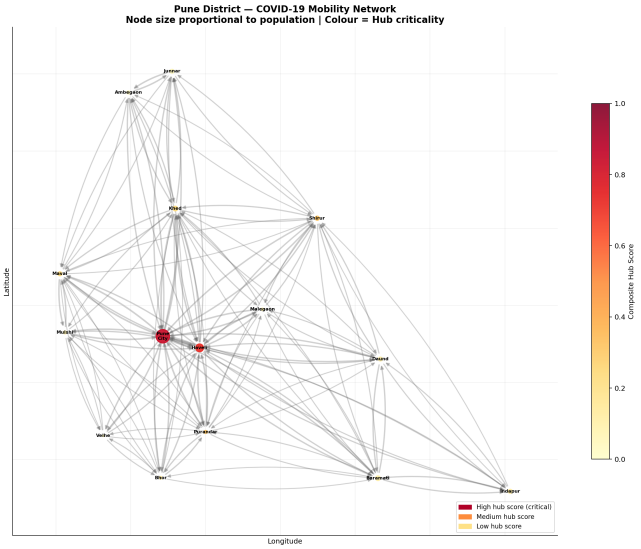


Figure 15: Pune District mobility network. Nodes at geographic coordinates; size ∝ population; colour ∝ composite hub score. Edge width ∝ $\log(F_{ij})$. The spoke-and-hub topology with Pune City and Haveli at the centre is immediately apparent.

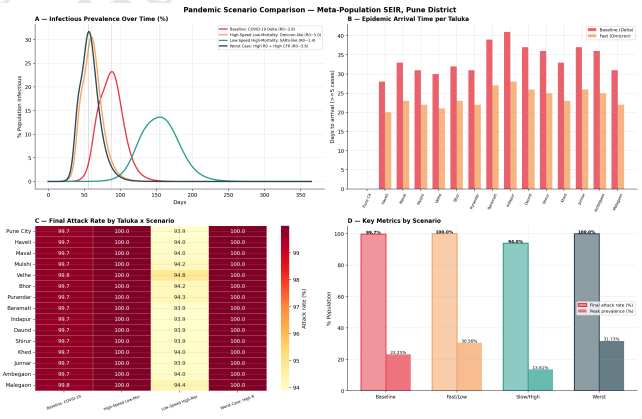


Figure 16: Metapopulation SEIR scenario comparison (four pathogen profiles). Key result: for all scenarios with $R_0 \geq 1.4$, final attack rate exceeds 93%—confirming that peak prevalence, not attack rate, is the actionable management target.

C SOFTWARE ENVIRONMENT

Python 3.10; key packages: `networkx==3.1`, `geopandas==0.13`, `python-louvain==0.16`, `scipy==1.11`, `folium==0.14`, `matplotlib==3.7`, `seaborn==0.12`, `scikit-learn==1.3`, `libpysal==4.7`, `esda==2.5`, `pymannkendall==1.4`.

Received January 2025; revised March 2025; accepted April 2025

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